Deal Document Intelligence Assistant

AI Solutions Internship — Project Plan

4–6 Weeks · Primarily Remote · Phase 1: Space Investments · Phase 2: AI Investments

INTERN	MENTOR	SPONSOR	DURATION
Aksel Rynearson	Greg Landegger	TWC Venture Team	4–6 Weeks

Project Phases

The project is structured in two phases. Phase 1 is the active internship scope. Phase 2 is a natural follow-on once the foundation is proven.

Phase 2 — AI Investments (Future)

- Dropbox archive of space investment pitch decks and business plans
- Sectors: launch vehicles, satellites, space infrastructure, defense, earth observation, in-space services
- Full 4–6 week build, evaluation, and demo
- Delivers the reusable technical foundation

- Dropbox archive of AI company pitch decks and business plans
- Sectors: AI infrastructure, foundation models, AI agents, vertical AI applications
- Extends the same pipeline to a new document corpus
- Same question types, new domain context

1. Core Deliverable

This project has one sharp, demonstrable goal:

A private, source-cited document intelligence assistant that makes the firm's space investment pitch decks and business plans searchable, understandable, and reusable.

The first version should not try to become a full investment intelligence platform or autonomous analyst. It should prove the intern can build a practical AI tool that ingests real documents, extracts useful information, retrieves relevant evidence, and answers questions with citations back to the source documents.

The same pipeline — with minimal changes — becomes Phase 2 when pointed at the AI investment archive.

2. Business Problem

The firm has accumulated a substantial archive of space investment pitch decks, business plans, and supporting materials. Valuable information is trapped inside PDFs, PowerPoints, and inconsistent folder structures — difficult to search, compare, or reuse.

Analysts and partners should be able to ask practical questions such as:

Example Question (Phase 1: Space)	Why It Matters
“Show all companies we reviewed in satellite communications.”	Compare a new opportunity against historical space deal flow.
“Find prior investments in launch vehicle or propulsion technology.”	Identify relevant precedents before evaluating a new launch startup.
“Which space companies mentioned revenue above $5M?”	Quickly filter by commercial traction without opening every deck.
“Compare space infrastructure opportunities across the archive.”	Support faster market mapping and comparable-company review.
“Which decks mention regulatory risk or launch licensing issues?”	Identify recurring diligence themes across the space portfolio.
“Summarize companies in earth observation and cite the source slides.”	Reduce manual document search and re-reading time.

The same question types apply directly to Phase 2 — simply replace “satellite communications” with “LLM infrastructure” or “AI agents.” The pipeline is identical; only the document corpus changes.

Today, answering these questions requires manual Dropbox search, file opening, skimming, and memory. This project eliminates that friction by creating a searchable, source-cited AI layer over a controlled document set.

3. Scope Philosophy

The project is managed in three tiers. Tier 1 must ship first. Tier 2 is the recommended internship target. Tier 3 is attempted only if the foundation is stable.

Tier	Description	Est. Hours	Recommendation
Tier 1: MVP	Dropbox ingestion, PDF/PPTX parsing, chunking, vector search, cited Q&A, simple interface, basic README	60–90 hrs	Appropriate if new to applied AI.
Tier 2: Recommended	Tier 1 plus company profiles, metadata filters, evaluation set, error logging, polished demo, sanitized GitHub, impact writeup	100–140 hrs	Official internship target.
Tier 3: Stretch	Tier 2 plus agentic memo workflow, hybrid search, OCR, duplicate detection, dashboard, larger corpus	150–220+ hrs	Only if core is working early.

The agentic layer is not removed — it is protected by sequencing. A polished, source-cited RAG system with one well-executed workflow is a stronger portfolio artifact than a half-finished agent built on unreliable retrieval.

4. Before Day 1 — Mentor Action Required

The project moves faster if the mentor prepares access, benchmark questions, and success criteria before the intern begins. The intern should not spend the first week waiting for permissions or guessing what the business actually needs.

Preparation Item	Description
Approved data folder	Create or identify a Dropbox folder with approved space investment documents. Start with 30–50 representative files.
Document mix	Include a mix of PDFs and PowerPoint files. Include a few image-heavy PDFs so parsing challenges surface early.
Access method	Provide Dropbox API access or an approved local export if API access is delayed.
Benchmark questions	Write 5–10 real questions an analyst might ask of the space investment archive. These become the evaluation benchmark.
Weekly meetings	Set a recurring weekly planning meeting (15–20 min) and weekly live demo (20–30 min).

Why Benchmark Questions Matter
The 5–10 benchmark questions are the single most important pre-start action.
They make the project concrete, give Aksel a realistic target from Day 1,
and create a way to measure improvement week over week.
Use real questions your team actually asks — not hypotheticals.

5. Week-by-Week Plan

The following plan assumes a 6-week project at 15–20 hours per week. If compressed to 4 weeks, see Section 6.

WEEK 1 ★ MUST SHIP	Data Foundation
Outcomes	• Create project repository, folder structure, .gitignore, .env.example, and initial README • Connect to the approved Dropbox folder or process an approved sample export

WEEK 1 ★ MUST SHIP	**Data Foundation**
	• Build a document inventory: file names, types, sizes, folder paths, and processing status • Extract text from a sample of PDFs and PowerPoint files • Identify and log hard cases: image-only PDFs, corrupted files, unusually large decks
Demo	*"I connected to the approved space investment archive, created a document inventory, extracted text from sample files, and identified the hard cases."*

WEEK 2 ★ MUST SHIP	**Basic RAG Pipeline**
Outcomes	• Implement text chunking strategy — preserve document name and slide/page number • Generate embeddings and store in a local vector database (Chroma or Pinecone free tier) • Connect retrieval to Claude to generate answers from retrieved source chunks • Run the mentor's benchmark questions — document what worked and what failed honestly
Demo	*"I can ask a natural-language question about our space investment archive and get a sourced answer. Here is how it performs on the 5 benchmark questions."*

WEEK 3 ★ MUST SHIP	**Retrieval Quality & Metadata**
Outcomes	• Compare at least two chunking approaches or retrieval settings — document the tradeoff • Add document-level metadata: company name, sector (launch, satellite, defense, etc.), date, stage • Implement filtered search: "show me only satellite companies" or "pre-revenue launch startups" • Re-run the same benchmark questions — show measurable improvement vs Week 2 • Identify the top retrieval failure modes and proposed fixes
Demo	*"Here are the same benchmark questions from Week 2. The answers are better, and here is exactly what changed."*

WEEK 4 ★ MUST SHIP	**Company Cards & Cited Q&A**
Outcomes	• Define company profile schema: name, sector, business model, stage, funding ask, key metrics, risks • Use LLM-assisted extraction to populate company cards with citations from source documents • Ensure all Q&A answers cite document name and page or slide number

WEEK 4 ★ MUST SHIP	Company Cards & Cited Q&A
	• Build a Streamlit or lightweight web interface for search, Q&A, and company-card viewing • Self-assess: is the foundation solid enough to add an agentic workflow in Weeks 5–6?
Demo	*"The system creates company cards for space investments and answers questions with citations the mentor can inspect."*

WEEK 5	Evaluation, User Feedback & Optional Agent
Outcomes	• Expand benchmark to 30–50 realistic space investment queries • Measure retrieval quality: does the right source appear in the top results? • Review answer groundedness: are generated answers supported by cited excerpts? • Have Greg or a colleague use the system without detailed guidance — document their feedback • Optional: begin the Due Diligence Briefing Agent workflow if the foundation is stable (see Section 7)
Demo	*"A real user tested the prototype on actual space investment questions. Here is what they could answer, where it failed, and what changed."*

WEEK 6	Polish, Document & Present
Outcomes	• Fix top issues from Week 5 user testing • Complete GitHub README with setup instructions, architecture diagram, and example outputs • Confirm no confidential documents, embeddings, API keys, or proprietary outputs are committed • Document final evaluation: metrics, failures, limitations, and next steps • Deliver 15–20 minute final presentation with live demo
Demo	*"End-to-end live demonstration across the space investment archive — what it solves, how it works, how well it performs, where it fails, and what comes next."*

6. Four-Week Compressed Plan

If the project needs to be completed in four weeks, focus on the MVP and avoid stretch features. Target 60–80 hours.

Week	Hours	Focus	Expected Output
Week 1	15–20	Data access, inventory,	Approved corpus and text extraction

Week	Hours	Focus	Expected Output
		PDF/PPTX parsing	pipeline
Week 2	15–20	Chunking, embeddings, vector search	Natural-language search with source snippets
Week 3	15–20	Cited Q&A and simple interface	Usable prototype that answers questions with citations
Week 4	15–20	Evaluation, documentation, final demo	Basic evaluation report, README, demo, and next-step plan

The compressed version should not include OCR, duplicate detection, dashboards, production deployment, or agentic workflows unless the core MVP finishes early.

7. The Agentic Feature — Done Right

If the foundation is solid by Week 5, implement one controlled agentic workflow: a Due Diligence Briefing Agent. This is a structured multi-step workflow, not an open-ended autonomous agent.

Due Diligence Briefing Agent — Space Investment
INPUT: A company name, sector (e.g. “launch vehicles”), or selected company card
STEP 1: Search the internal archive for all related space investment documents and extract key facts
STEP 2: Pull structured company-card data: sector, stage, funding ask, key metrics, and risks
STEP 3: Identify missing information and likely diligence questions for a space investment context
STEP 4: Search the web for current news, recent funding rounds, and competitive context
OUTPUT: A one-page internal briefing that distinguishes sourced facts from assumptions
The agent should NOT make investment recommendations, contact external parties,
submit forms, or take any action outside the approved research workflow.

This demonstrates multi-step AI workflow design while remaining safe and explainable. It maps directly to real agentic products and answers the question a USC Viterbi interviewer will ask: “What can this system do that a junior analyst doing manual research cannot?”

The same agent, pointed at the AI investment corpus in Phase 2, requires minimal modification.

8. Roles & Responsibilities

Greg Landegger — Mentor	Aksel Rynearson — Intern
• Provide approved Dropbox access or sample export before Day 1 • Write 5–10 benchmark questions from real analyst workflows • Participate in weekly planning and demo sessions • Give feedback on usefulness and business relevance • Help unblock access and logistics issues • Provide final evaluation and reference if appropriate	• Build working prototypes independently • Make all technical decisions and document the tradeoffs • Maintain a clean repository with regular commit history • Maintain an engineering journal throughout the project • Show live progress, including failures and limitations • Own the final demo, documentation, and project narrative • Ensure no confidential data or secrets are exposed in public materials

The mentor is not expected to review code, design architecture, troubleshoot implementation, or select tools. The independence is part of the learning experience.

9. Working Cadence

Meeting	Duration	Purpose
Weekly planning	15–20 min	Review prior progress, set weekly priorities, identify blockers, adjust scope.
Weekly demo	20–30 min	Live functionality demo, explain what worked, show one failure, preview next steps.
Mid-project stakeholder review	30 min	A second user tests the prototype and gives feedback on usefulness.
Final presentation	15–20 min	Business problem, architecture, live demo, evaluation, lessons learned, roadmap.

Demo Norm: Show a Failure Every Week
Every weekly demo should include at least one thing that did not work as expected.
This is not a sign of poor performance — it is a sign of honest engineering.
Real engineers spend more time debugging than building. Demonstrating that maturity
is one of the most impressive things a junior engineer can show a future employer.

10. Engineering Journal

Aksel should maintain a brief engineering journal for each significant technical decision. Short, honest entries are better than polished summaries written after the fact.

Field	Prompt
Date	When was the decision made?
Objective	What was I trying to accomplish?
What I built or tried	What implementation or experiment did I run?
Challenge encountered	What did not work as expected?
Alternatives considered	What other approaches did I consider?
Decision made	What did I choose, and why?
Result	What improved, broke, or remained unresolved?
Next step	What will I do next?

11. GitHub Repository Standards

The GitHub repository is a portfolio artifact. Future employers and USC Viterbi faculty may review it. Before anything is made public, Greg should review the repository to confirm no confidential data or proprietary outputs are included.

Standard	Requirement
Commit history	Commit regularly with meaningful messages that explain what changed and why.
README	Project description, architecture, setup instructions, example queries, limitations, and lessons learned.
Architecture diagram	Show ingestion, parsing, chunking, vector store, retrieval, LLM generation, UI, and evaluation.
Sample data	Use synthetic or public sample documents only. Never commit real deal documents.
Secrets	Never commit API keys, Dropbox tokens, .env files, credentials, logs, embeddings, or proprietary documents.
Code comments	Comment important design decisions and non-obvious tradeoffs.
Evaluation report	Benchmark questions, results, failure modes, and improvement notes.

12. Success Metrics

The project is evaluated on usefulness, reliability, and learning quality. A visually impressive demo is not enough if retrieval is unreliable or answers cannot be verified.

Metric	What It Measures	Target
Benchmark questions	Can the system answer real space investment questions?	Run same questions Weeks 2, 3, and 4 to show improvement.
Documents processed	How reliable is ingestion and parsing?	80%+ of pilot documents processed; failures logged.
Retrieval quality	Does the right source appear in top results?	75%+ hit rate@5 on 30–50 evaluation questions (Tier 2).
Answer groundedness	Are generated answers supported by citations?	80%+ of reviewed answers supported by cited excerpts.
Company profile quality	Are extracted fields accurate enough to use?	80%+ accuracy for core fields after review.
User test	Can a non-technical user use the tool?	At least one real user test by Week 5.
Time-to-answer	Does the system reduce manual search time?	Demonstrate faster answers for representative questions.
GitHub quality	Is the project understandable to an outside reviewer?	Clean README, architecture diagram, no confidential content.
Engineering maturity	Does Aksel document tradeoffs and failures?	Engineering journal includes real decisions and lessons learned.

13. Final Deliverables

Project Artifacts	Presentation & Handoff
• Working prototype — live, demo-ready for search, cited Q&A, and company profile viewing • Document processing pipeline: ingestion, parsing, chunking, embedding, and indexing • Company profile dataset for the space investment pilot corpus, with citations • Evaluation report: benchmark results, retrieval quality, groundedness, known limitations • Engineering journal covering key decisions throughout the project	• Sanitized GitHub repository: code, README, architecture diagram, sample data, setup instructions • Final presentation: 15–20 minutes with live demo, architecture, results, lessons learned, next steps • Phase 2 handoff note: what would need to change to point the pipeline at the AI investment corpus • Impact estimate: time saved, improved archive usability, and recommended next steps for the firm

14. Final Presentation Structure

The final presentation should be concise and demo-driven. It should show that Aksel understands both the business problem and the technical system.

Section	Suggested Content
Business problem	Why the space investment archive is hard to use today and why it matters.
Project goal	The specific deliverable and what was intentionally out of scope.
Architecture	How Dropbox ingestion, parsing, chunking, embeddings, retrieval, LLM answers, and UI fit together.
Live demo	Search the archive, ask a space investment question, inspect citations, view a company card, optionally generate a diligence brief.
Evaluation	Benchmark questions, retrieval performance, groundedness review, and user feedback.
Failures & lessons	What did not work and what changed because of it.
Impact & Phase 2	Time saved, archive usability improvements, and what it would take to extend to the AI investment corpus.
Portfolio summary	What can be shared publicly and what remains confidential.

15. Resume & Portfolio Positioning

The project should be described as a real applied AI system, not an API experiment. The strongest story is that Aksel built a working pipeline, handled messy documents, improved retrieval, evaluated quality, and delivered something a business user could test.

Suggested Resume Bullet (complete after project — replace placeholders with real numbers)
Built a secure AI-powered deal document intelligence prototype for a space investment
research use case, integrating Dropbox ingestion, PDF/PPTX parsing, vector search,
source-cited Q&A, and structured company metadata extraction; measured retrieval quality
against real analyst benchmark questions and delivered a sanitized GitHub portfolio demo.
Shorter version:
Developed a RAG-based document intelligence application that transformed space investment
pitch decks and business plans into a searchable, source-cited research assistant with
company profiles, evaluation metrics, and a polished demo package.

For a freshman entering USC Viterbi CS, this project signals three things that matter most:

- Systems thinking — not just “I used an API” but “I designed a pipeline with these tradeoffs.”
- Real deployment — something a non-technical business user actually tested and got value from.
- Honest evaluation — showing what didn’t work and why demonstrates engineering maturity beyond most freshman-year projects.

16. Final Mentor Guidance

This is a strong summer project if it remains focused. The recommended official target is Tier 2: 100–120 hours over 5–6 weeks.

- Push for a working, demonstrable core system before allowing advanced features.
- If Aksel finishes the core system early, add the controlled due diligence briefing workflow.
- If the core system is unstable at Week 4, cut the agentic feature and focus on retrieval quality, citations, evaluation, and documentation.
- The best outcome is not a broad platform — it is a reliable prototype that answers real space investment questions, cites its sources, and gives Aksel a credible applied AI project story.
- Phase 2 (AI investments) should only begin after Phase 1 is fully delivered, documented, and reviewed.

This project, done well, is stronger than most sophomore-year internship projects. Keep scope tight in Weeks 1–4 so there is something real to build on.